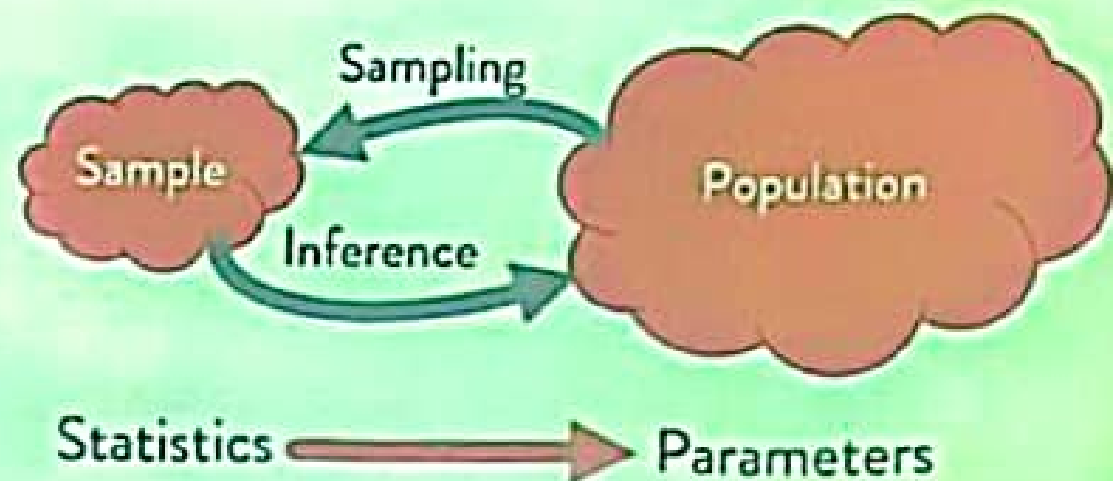
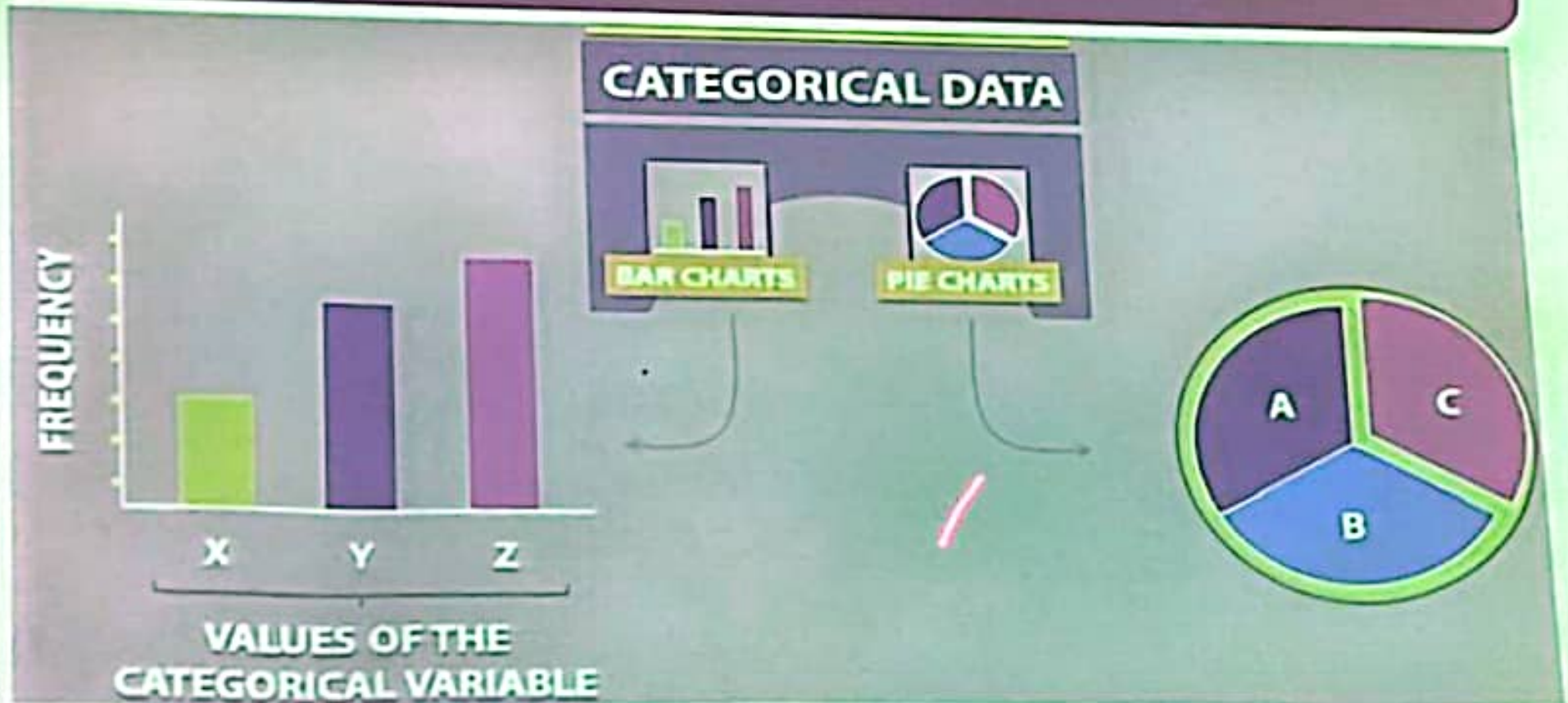


Parameters and Statistics

- **Statistics** are numbers that describe data of a sample.
- **Parameters** are numbers that describe data of a population.
- We use sample statistics to estimate population parameters.



Graphs for Qualitative Data



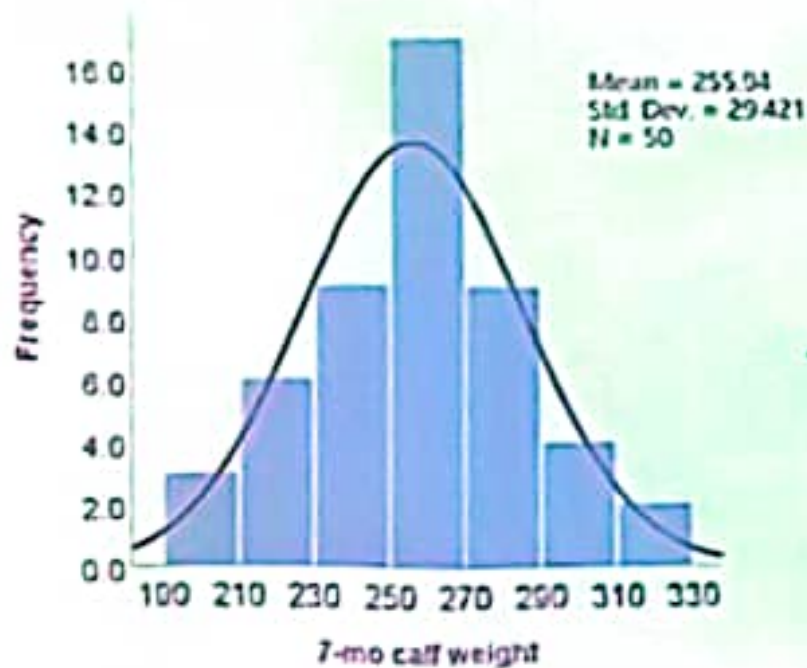
Quantitative Data

Frequency Tables
and Histograms

Stem and Leaf plot

Boxplot

What do histograms tell us?



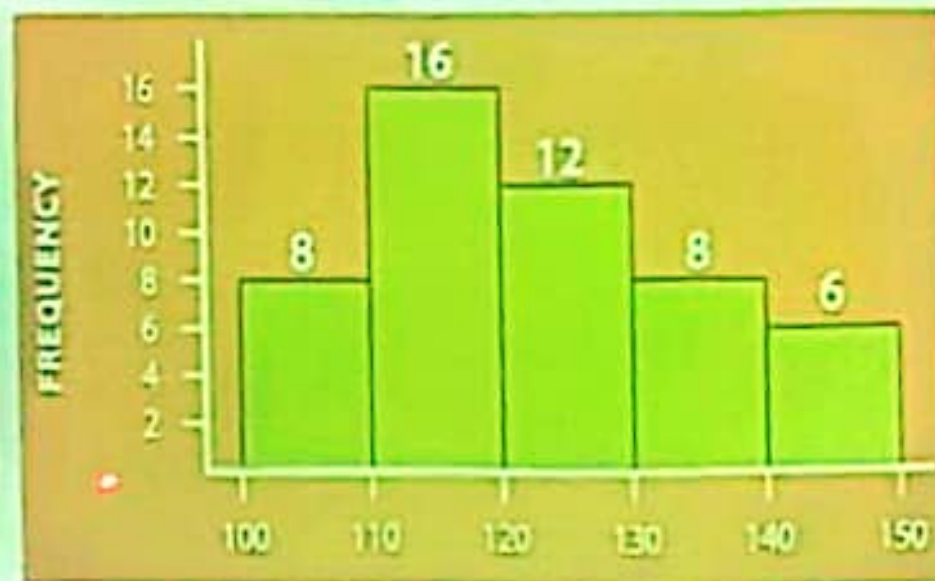
It shows basic information about the data
e.g.

- Center and variation (**spread**) of data
- **Shape**: If the left and the right sides are nearly similar, then the data are said to be symmetric about the mean, i.e., normally distributed.

Quantitative Data

Frequency Tables and Histograms

WEIGHT	FREQUENCY
100 – 110	8
110 – 120	16
120 – 130	12
130 – 140	8
140 – 150	6



Quantitative Data: Stem and Leaf plot

Each value is divided into two parts, "Stem" and "Leaf".

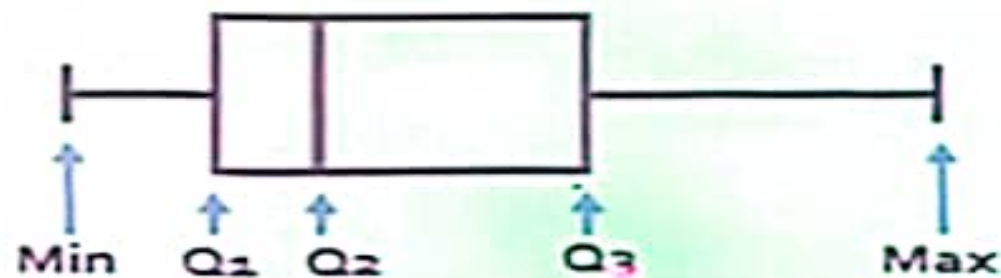
STEM	LEAF
REFERS TO ALL OF THE OTHER NUMBERS EXCEPT THE LAST NUMBER	REFERS TO THE VERY LAST NUMBER

117

117

11 | 7

Quantitative Data: The Boxplot (the Five-Number Summary)



"Stem and Leaf" plot of 7-month weights calves

Stem	Leaf
19	0 7
20	9
21	0 1 5
22	4 7 7
23	0 1 6 8
24	1 1 4 7 9
25	0 1 1 2 4 5 5 5 6 6
26	2 5 5 6 6 8 9
27	1 1 2 3 5 7
28	8 9 9
29	1 2
30	0 3
31	8
32	5

Legend 19|0 represents 190

- Similar to the histogram with the advantage that it shows **all values** of the data.
- It is easy to find the
 - range ($325 - 190 = 135$),
 - mode (255),
 - median (255)
- Useful information e.g. calves weighing ≥ 300 kg (4), and calves weighing < 220 kg (6).

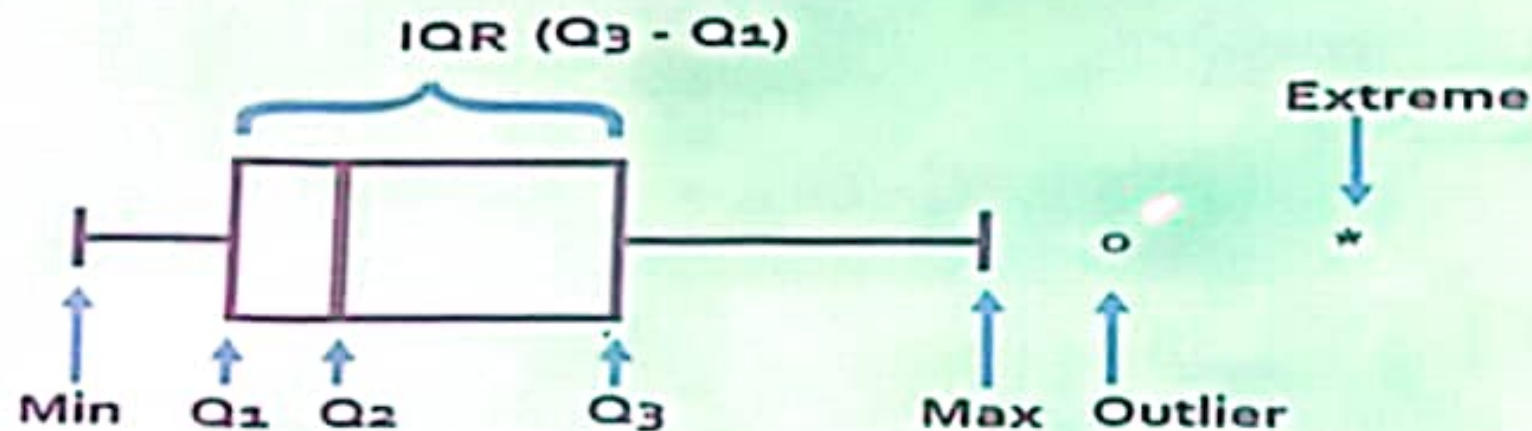


A part of a population selected for analysis.

A good sample (unbiased sample)

- is representative of the entire population
- gives everyone an equal chance of being chosen

Quantitative Data: The Boxplot (the Five-Number Summary)



Simple Random Sampling

- Prepare a list of all units

Table 1 from Appendix A: Random Numbers

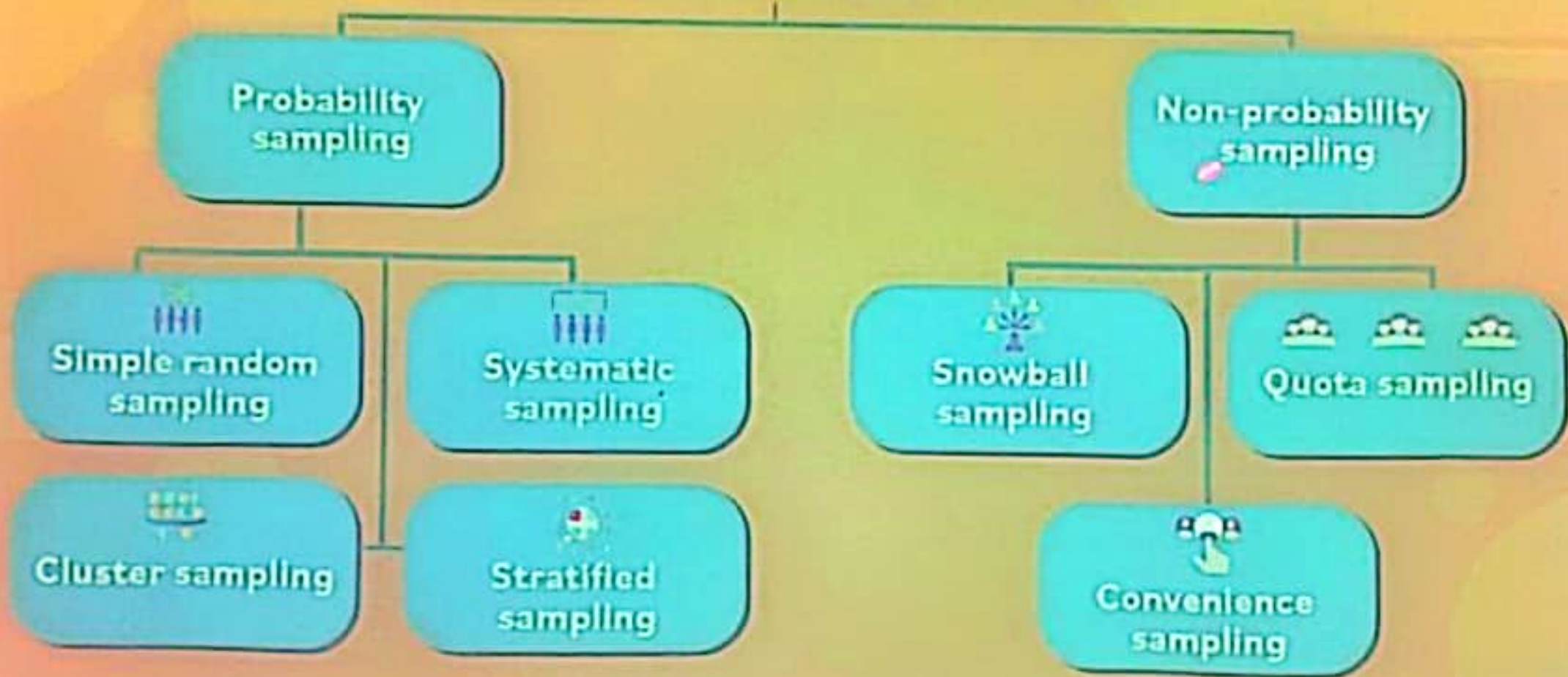
Table 1							
Random Numbers							
Row Number	Column Number						
	01-05	06-10	11-15	16-20	21-25	26-30	31-35
01	89392	23212	74483	36590	25956	36544	685
02	61458	7639	96252	95649	73727	33912	726
03	115	4197	81962	48443	90360	26480	733
04	27	04429	31308	02241	01698	19191	189
05	29	59109	88976	46845	28329	47460	889
06	19032	93458	42161	26099	09419	89073	829
07	59761	55212	33360	68751	86737	79743	852
08	46827	25906	64708	20307	78423	15910	865
09	24040	66449	32353	83668	13874	86741	813
10	98144	96372	50277	15571	82261	68628	314
11	14228	17930	30118	06438	49660	65189	628
12	55366	31057	96065	14791	62426	02957	855
13	96101	30846	35526	98369	73634	79304	989
14	38152	55474	30153	26525	83647	31988	821



- OR a random number generator, calculator, excel ...

See page 6

Sampling Methods



Systematic Random Sampling - Example

- Suppose you want to sample 5 cows of 200 in the faculty farm to determine antibody level to a disease.
- K (sampling interval) = $K = \frac{200}{5} = 40$
- Choose a random start between 1 and 40, say 15. Add 40 systematically.
- Selected cows are 15th, 55th, 95th, 135th, and 175th.

Cluster sampling

- If the population is spread over a **large area**, collecting the sample may be very time- and money-consuming

